SKTC 1623

Thermodynamics

Chapter 1: Introduction and Basic Concepts

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Overview

- **Thermodynamics** can be defined as the science of *energy*.
- Energy can be viewed as the ability to cause changes.
- Thermodynamics is broadly interpreted to include all aspects of **energy** and **energy transformations**, including power production, refrigeration, and relationships among the properties of matter.
- One of the most fundamental laws of nature is the **Conservation of Energy Principle**
  - energy can change from one form to another but the total amount of energy remains constant.
  - energy cannot be created or destroyed.
- The **first law of thermodynamics** is simply an expression of the conservation of energy principle
  - asserts that energy is a thermodynamic property.

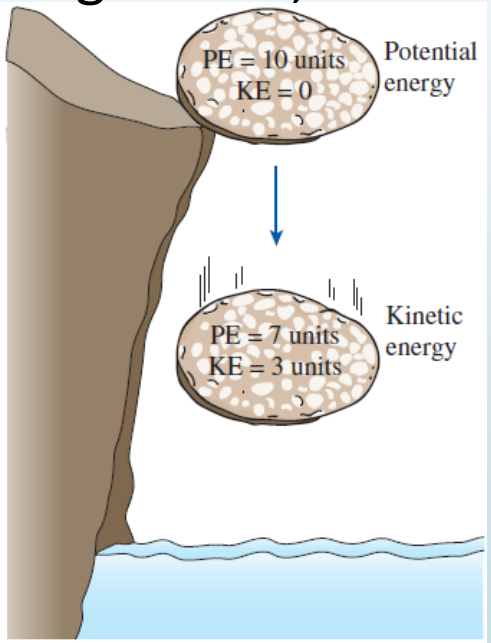

FIGURE 1–1
Energy cannot be created or destroyed; it can only change forms (the first law).

Energy (2)

- The **second law of thermodynamics** asserts that energy has quality as well as quantity, and actual processes occur in the direction of decreasing quality of energy.
- **Classical thermodynamics:** A macroscopic approach to the study of thermodynamics that does not require a knowledge of the behavior of individual particles.
  - It provides a direct and easy way to the solution of engineering problems and it is used in this text.
- **Statistical thermodynamics:** A microscopic approach, based on the average behavior of large groups of individual particles.

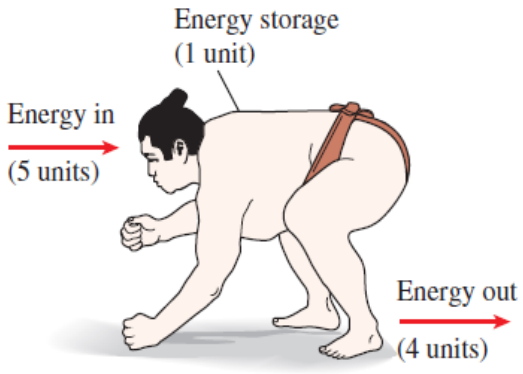

FIGURE 1–2
Conservation of energy principle for the human body.

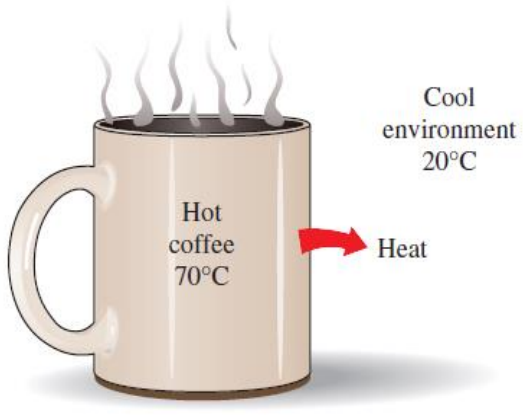

FIGURE 1–3
Heat flows in the direction of decreasing temperature.

Application Areas of Thermodynamics

All activities in nature involve some interaction between energy and matter; thus, it is hard to imagine an area that does not relate to thermodynamics in some manner.

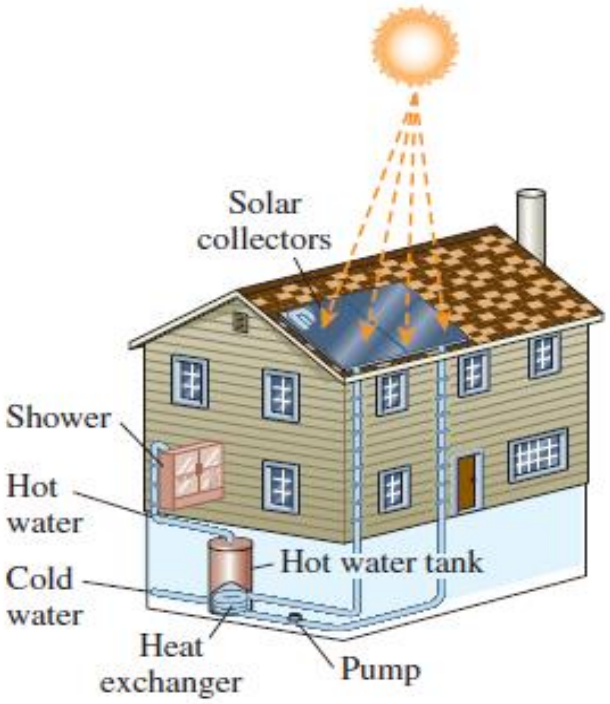

FIGURE 1–4
The design of many engineering systems, such as this solar hot water system, involves thermodynamics.

Refrigerator
© McGraw-Hill Education, Jill Braaten

Boats
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Aircraft and spacecraft
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Power plants
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Human body

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Cars

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Wind turbines

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Food processing

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A piping network in an industrial facility.

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1.2 Dimensions and Units

- Any physical quantity can be characterized by **dimensions**.
- The magnitudes assigned to the dimensions are called **units**.
- Some basic dimensions such as mass *m*, length *L*, time *t*, and temperature *T* are selected as **primary** or **fundamental dimensions**, while others such as velocity *V*, energy *E*, and volume *V* are expressed in terms of the primary dimensions and are called **secondary dimensions**, or **derived dimensions**.
- **Metric SI system**: A simple and logical system based on a decimal relationship between the various units.
- **English system**: It has no apparent systematic numerical base, and various units in this system are related to each other rather arbitrarily.

TABLE 1–1

The seven fundamental (or primary) dimensions and their units in SI

Dimension	Unit
Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Temperature	kelvin (K)
Electric current	ampere (A)
Amount of light	candela (cd)
Amount of matter	mole (mol)

TABLE 1–2

Standard prefixes in SI units

Multiple	Prefix
10^{24}	yotta, Y
10^{21}	zetta, Z
10^{18}	exa, E
10^{15}	peta, P
10^{12}	tera, T
10^{9}	giga, G
10^{6}	mega, M
10^{3}	kilo, k
10^{2}	hecto, h
10^{1}	deka, da
10^{-1}	deci, d
10^{-2}	centi, c
10^{-3}	milli, m
10^{-6}	micro, μ
10^{-9}	nano, n
10^{-12}	pico, p
10^{-15}	femto, f
10^{-18}	atto, a
10^{-21}	zepto, z
10^{-24}	yocto, y

1.2 Dimensions and Units (2)

- **Metric SI** (from *Le Système International d' Unités*), which is also known as the *International System*, is based on six fundamental dimensions. Their units, adopted in 1954 at the Tenth General Conference of Weights and Measures, are: *meter* (m) for length, *kilogram* (kg) for mass, *second* (s) for time, *ampere* (A) for electric current, *degree Kelvin* (K) for temperature, *candela* (cd) for luminous intensity (amount of light), and *mole* (mol) for the amount of matter.
- **English system**, which is also known as the *United States Customary System* (USCS), has the respective units the pound-mass (lbm), foot (ft), and second (s). The pound symbol *lb* is actually the abbreviation of *libra,* which was the ancient Roman unit of weight.

Some SI and English Units

$$1 \text{ lbm} = 0.45359 \text{ kg}$$

$$1 \text{ ft} = 0.3048 \text{ m}$$

$$\text{Force} = (\text{Mass})(\text{Acceleration})$$

$$F = ma$$

$$1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2$$

$$1 \text{ lbf} = 32.174 \text{ lbm}\cdot\text{ft/s}^2$$

Work = Force × Distance

1 J = 1 N·m

1 cal = 4.1868 J

1 Btu = 1.0551 kJ

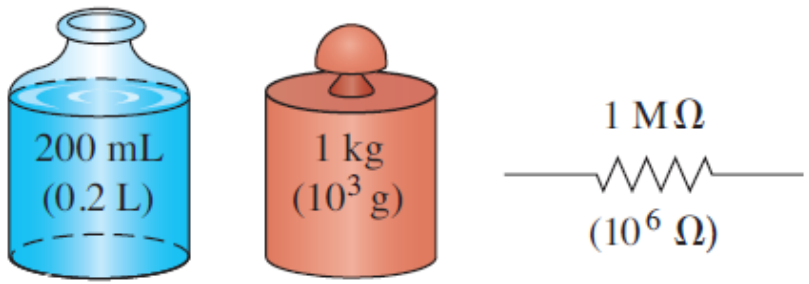


FIGURE 1–6
The SI unit prefixes are used in all branches of engineering.

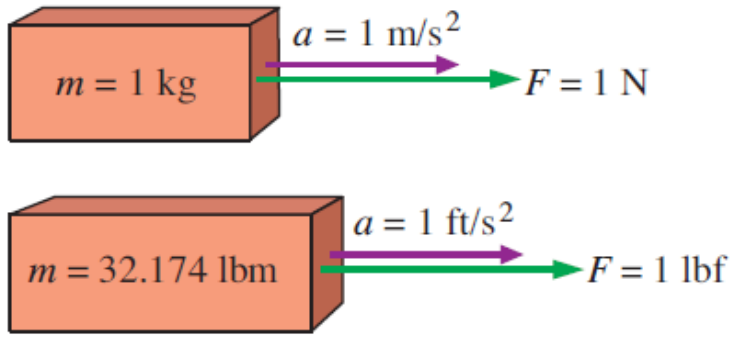


FIGURE 1–7
The definition of the force units.

Weight

$$W = mg \quad (\mathrm{N})$$

W = weight
m = mass
g = gravitational acceleration

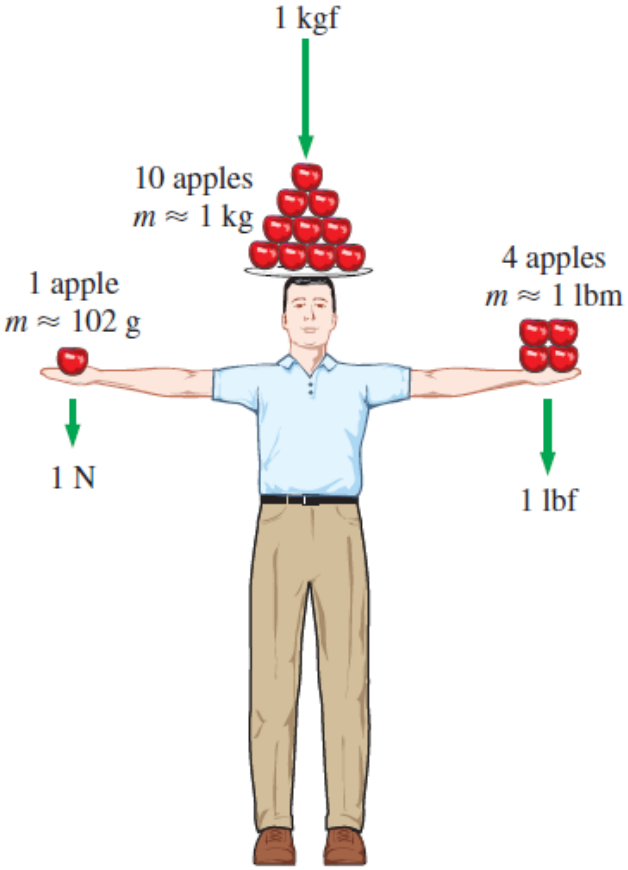


FIGURE 1–8
The relative magnitudes of the force units newton (N), kilogram-force (kgf) and pound-force (lbf).

FIGURE 1–9
A body weighing 66 kgf on earth will weigh only 11 kgf on the moon.

Specific weight γ: The weight of a unit volume of a substance.

$$\gamma = \rho g$$

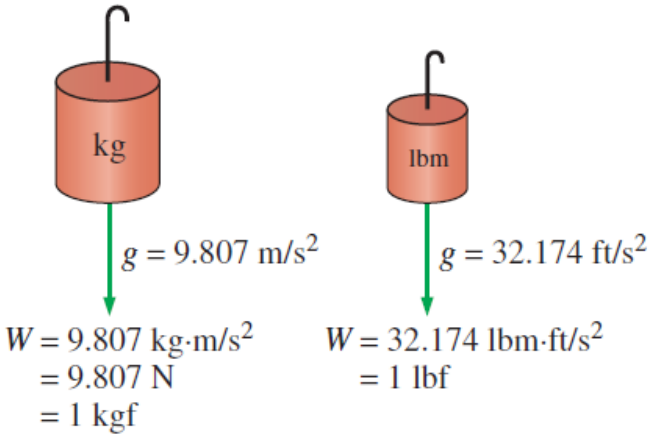


FIGURE 1–10
The weight of a unit mass at sea level.

FIGURE 1–11
A typical match yields about one Btu (or one kJ) of energy if completely burned.

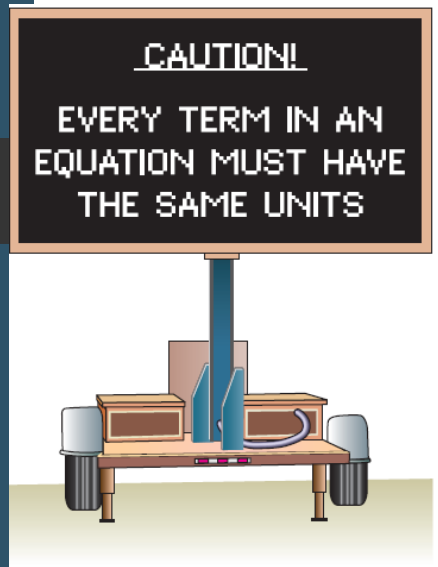


FIGURE
Always check the units in your calculations.

Dimensional homogeneity

Dimensionally homogeneous means that every term in an equation must have the same unit.

To make sure that all terms in an engineering equation have the same units is the simplest error check one can perform.

Unity Conversion Ratios

All nonprimary units (secondary units) can be formed by combinations of primary units.

Force units, for example, can be expressed as

$$\mathrm{N} = \mathrm{kg}\,\frac{\mathrm{m}}{\mathrm{s}^2} \quad \text{and} \quad \mathrm{lbf} = 32.174\ \mathrm{lbm}\,\frac{\mathrm{ft}}{\mathrm{s}^2}$$

They can also be expressed more conveniently as **unity conversion ratios** as

$$\frac{\mathrm{N}}{\mathrm{kg}\cdot\mathrm{m/s}^2} = 1 \quad \text{and} \quad \frac{\mathrm{lbf}}{32.174\ \mathrm{lbm}\cdot\mathrm{ft/s}^2} = 1$$

Unity conversion ratios are identically equal to 1 and are unitless, and thus such ratios can be inserted conveniently into any calculation to properly convert units.

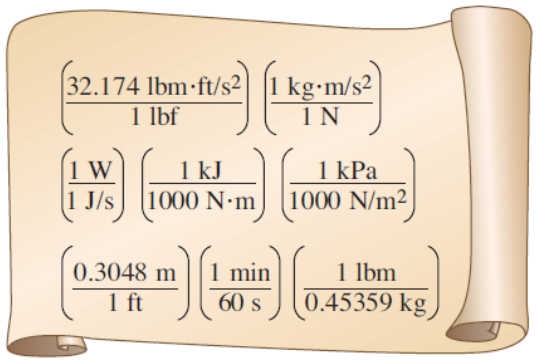


FIGURE 1–15
Every unity conversion ratio (as well as its inverse) is exactly equal to one. Shown here are a few commonly used unity conversion ratios.

1.3 Systems and Control Volumes

- A **System** is defined as a *quantity of matter or a region in space chosen for study.*
- **Surrounding** is the mass or region outside the thermodynamic system.
- B**oundary** is the real or imaginary surface that separates the system from its surroundings.
  - The boundary of a system can be ***fixed*** or ***movable***.
  - Systems may be considered to be ***closed*** or ***open,*** depending on whether a fixed mass or a fixed volume in space is chosen for study.

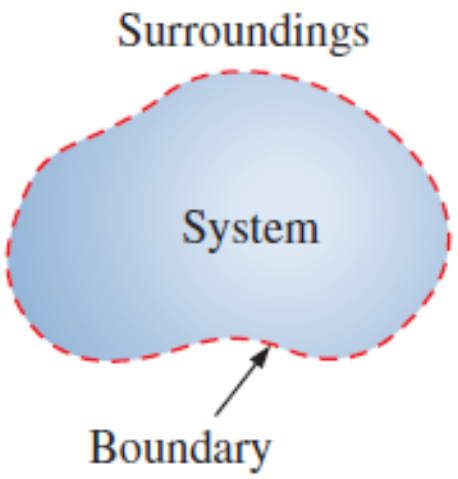


FIGURE 1–18
System, surroundings, and boundary.

1.3 Systems and Control Volumes (2)

- A **Closed system** (also known as a **control mass**) consists of a fixed amount of mass, and no mass can cross its boundary.
  - But energy, in the form of heat or work, can cross the boundary.
  - **Isolated system** is a closed system in which energy is also not allowed to cross the boundary.

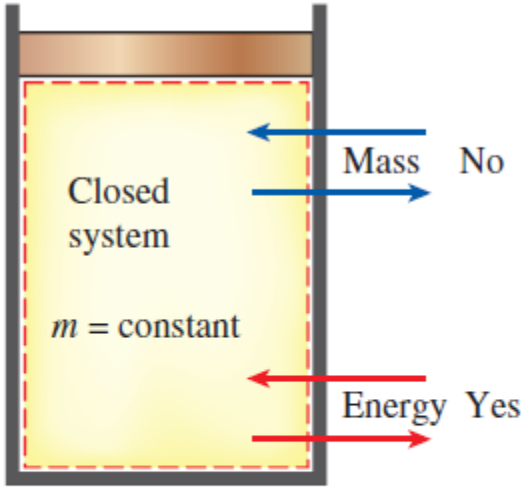


FIGURE 1–19
Mass cannot cross the boundaries of a closed system, but energy can.

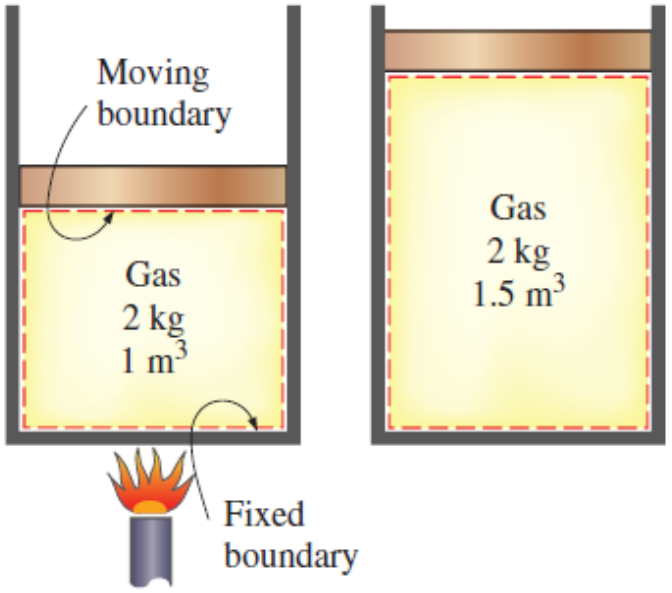
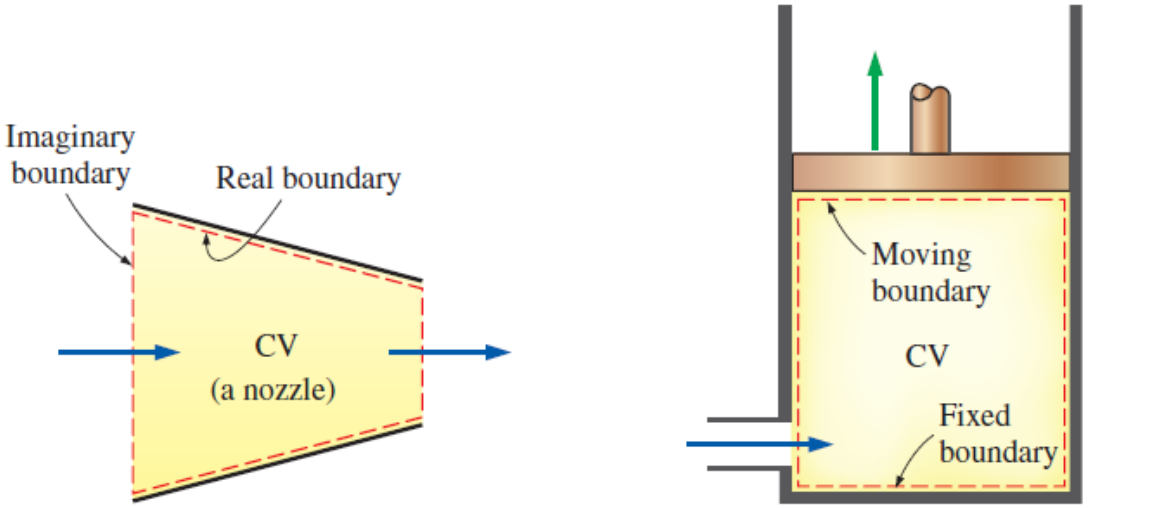


FIGURE 1–20
A closed system with a moving boundary.

1.3 Systems and Control Volumes (3)

- An **open** system, or a **control volume** is any arbitrary region in space through which mass and energy can pass across the boundary.
  - Usually encloses a device that involves mass flow such as a compressor, turbine, or nozzle.
  - A control volume may also involve heat and work interactions just as a closed system, in addition to mass interaction.

FIGURE 1–22
An open system (a control volume) with one inlet and one exit.



(*a*) A control volume (CV) with real and imaginary boundaries

(*b*) A control volume (CV) with fixed and moving boundaries as well as real and imaginary boundaries

A control volume can involve fixed, moving, real, and imaginary boundaries.

1.4 Properties of a System

- **Property** is any characteristic of a system**.**
  - Common: pressure P, temperature T, volume V, and mass m.
  - Others: viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electric resistivity, velocity and elevation.
  - Properties are considered to be either intensive or extensive.
- **Intensive properties**
  - Values independent of mass (the size of a system) - temperature, pressure, and density.
- **Extensive properties**
  - Values depend on the size—or extent—of the system.
  - Mass *m*, volume *V*, and total energy *E* are some examples of extensive properties.
- **Specific properties:** Extensive properties per unit mass.

$(v = V/m)$ $(e = E/m)$.

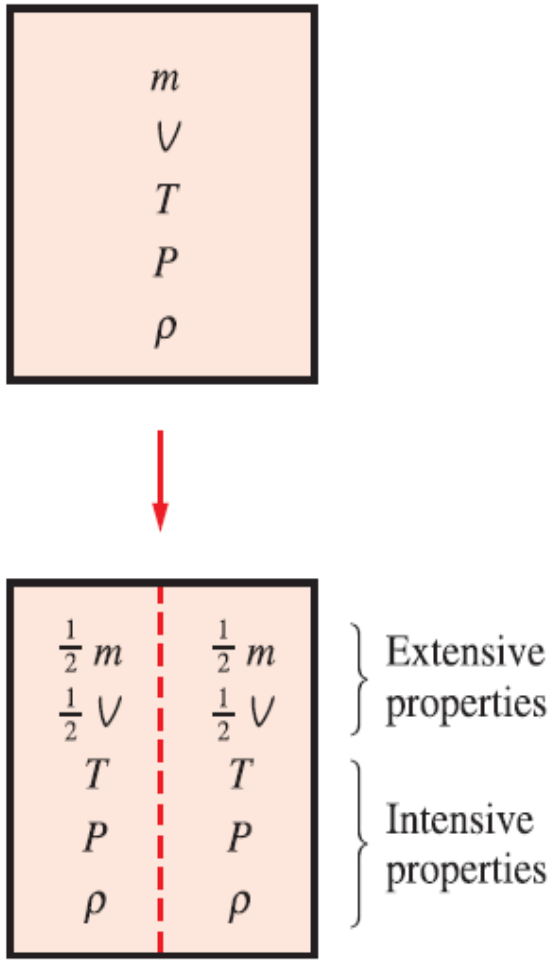


FIGURE 1–23
Criterion to differentiate intensive and extensive properties.

Continuum

- Matter is made up of atoms that are widely spaced in the gas phase.
- Yet it is very convenient to disregard the atomic nature of a substance and view it as a continuous, homogeneous matter with no holes, that is, a **continuum**.
- The continuum idealization allows us to
  - treat properties as point functions and
  - assume the properties vary continually in space with no jump discontinuities.
- This idealization is valid as long as the size of the system we deal with is large relative to the space between the molecules.
- In this text we will limit our consideration to substances that can be modeled as a continuum.

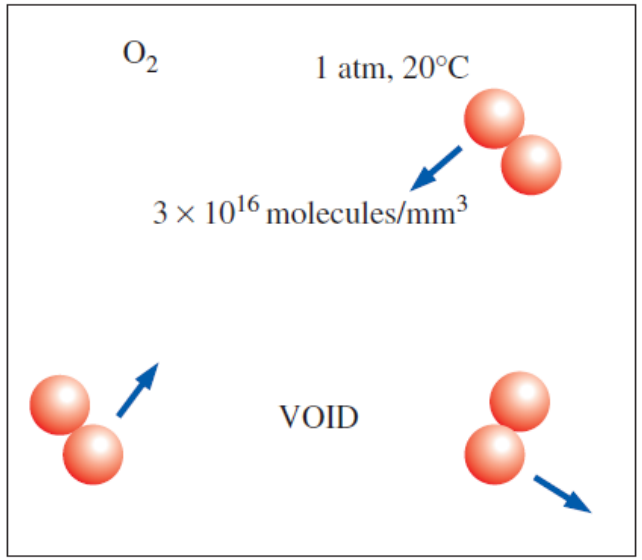


FIGURE 1–24
Despite the relatively large gaps between molecules, a gas can usually be treated as a continuum because of the very large number of molecules even in an extremely small volume.

1.5 Density and Specific Gravity

$V = 12\ \text{m}^3$
$m = 3\ \text{kg}$

$\rho = 0.25\ \text{kg/m}^3$

$v = \frac{1}{\rho} = 4\ \text{m}^3/\text{kg}$

- Density is mass per unit volume, $\rho = \frac{m}{V} \qquad (\text{kg/m}^3)$
- Specific volume is volume per unit mass, $v = \frac{V}{m} = \frac{1}{\rho}$
- **Specific gravity**: The ratio of the density of a substance to the density of some standard substance at a specified temperature (usually water at 4°C).

$$\text{SG} = \frac{\rho}{\rho_{H_2O}}$$

- **Specific weight**: The weight of a unit volume of a substance.

$$\gamma_s = \rho g \qquad (\text{N/m}^3)$$

TABLE 1–3

Specific gravities of some substances at 0°C

Substance	SG
Water	1.0
Blood	1.05
Seawater	1.025
Gasoline	0.7
Ethyl alcohol	0.79
Mercury	13.6
Wood	0.3–0.9
Gold	19.2
Bones	1.7–2.0
Ice	0.92
Air (at 1 atm)	0.0013

1.6 State and Equilibrium

- Consider a System not undergoing any change.
- At this point, all the properties can be measured or calculated throughout the entire system.
- This gives a set of properties that completely describes the condition of that system, or the **State** of the system

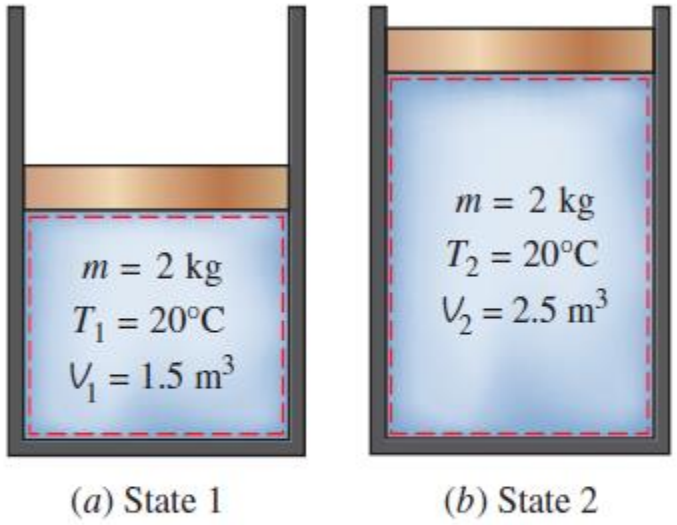
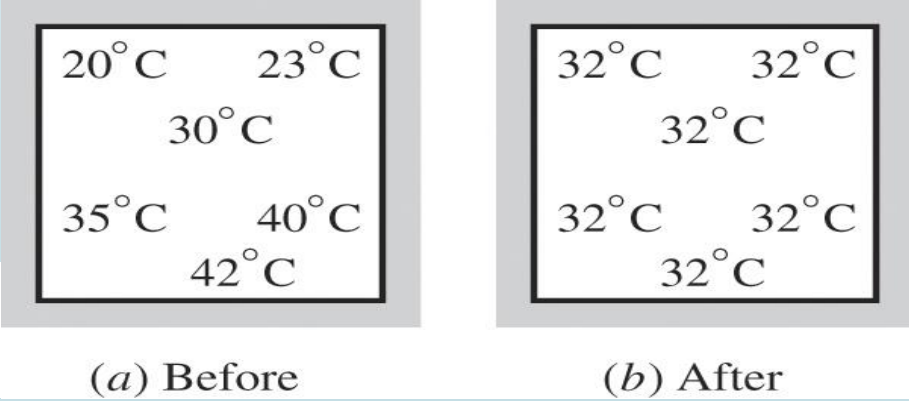


FIGURE 1–26
A system at two different states.

1.6 State and Equilibrium (2)

- **Equilibrium** implies a state of balance.
  - there are no unbalanced potentials (or driving forces) within the system.
- **Thermal equilibrium** means that the temperature is the same throughout the entire system.
- **Mechanical equilibrium:** If there is no change in pressure at any point of the system with time.
- **Phase equilibrium:** If a system involves two phases and when the mass of each phase reaches an equilibrium level and stays there.
- **Chemical equilibrium** is established in a system when its chemical composition does not change with time.



A closed system reaching thermal equilibrium

The State Postulate

- The number of properties required to fix the state of a system is given by the **state postulate**:
  - *The state of a simple compressible system is completely specified by two independent, intensive properties.*
- **Simple compressible system:** If a system involves no electrical, magnetic, gravitational, motion, and surface tension effects.
- Two properties are independent if one property can be varied while the other one is held constant (eg. T and *v*, T and P (for single-phase system)
  - Note that Pressure, P and temperature, T are not independent, if the system contains two phases

FIGURE 1–28
The state of nitrogen is fixed by two independent, intensive properties.

1.7 Processes and Cycles

- **Process**: Any change that a system undergoes from one equilibrium state to another.
- **Path**: The series of states through which a system passes during a process.
  - To describe a process completely, one should specify the initial and final states, as well as the path it follows, and the interactions with the surroundings.
- **Quasistatic or quasi-equilibrium process:**
  - When a process proceeds in such a manner that the system remains infinitesimally close to an equilibrium state at all times.

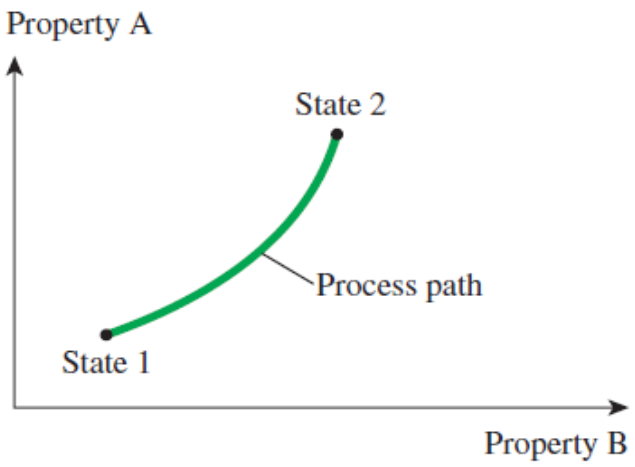


FIGURE 1–29
A process between states 1 and 2 and the process path.

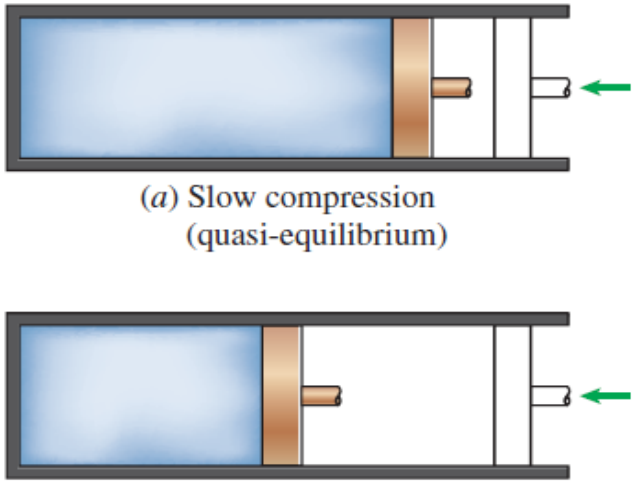


FIGURE 1–30
Quasi-equilibrium and nonquasi-equilibrium compression processes.

1.7 Processes and Cycles (2)

- Process diagrams plotted by employing thermodynamic properties as coordinates are very useful in visualizing the processes.
  - Some common properties used as coordinates are temperature *T*, pressure *P*, and volume *V* (or specific volume *v*).
- The prefix *iso-* is often used to designate a process for which a particular property remains constant.
  - **Isothermal process**: A process during which the temperature *T* remains constant.
  - **Isobaric process**: A process during which the pressure *P* remains constant.
  - **Isochoric** (isometric) **process** is a process during which the specific volume v remains constant.
- **Cycle** is a process, or series of processes, that allows a system to undergo state changes and returns the system to the initial state at the end of the process. That is, for a cycle the initial and final states are identical.

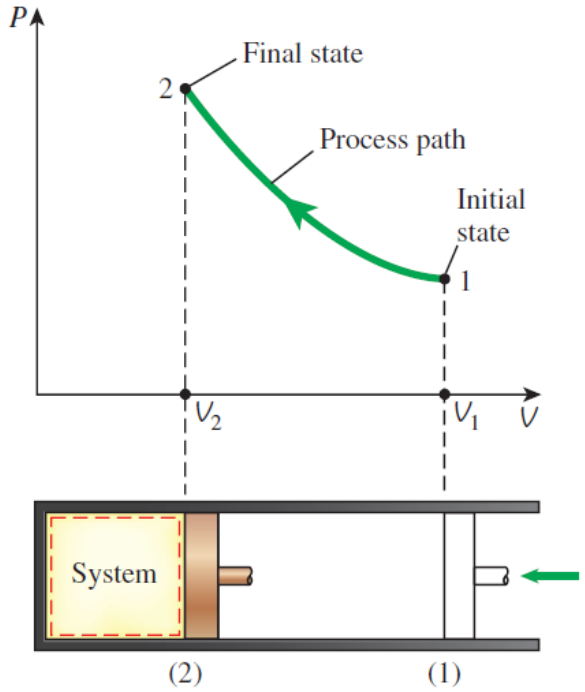


FIGURE 1–31
The *P-V* diagram of a compression process.

The Steady Flow Process

- **Steady** implies no change with time. The opposite of steady is unsteady, or transient.
  - A large number of engineering devices operate for long periods of time under the same conditions, and they are classified as steady-flow devices.
  - E.g.: turbines, pumps, boilers, condensers, and heat exchangers or power plants or refrigeration systems.
- **Steady-flow process** is defined as a process during which *a fluid flows through a control volume steadily*.
- The fluid properties can change from point to point within the control volume, but at any fixed point they remain the same during the entire process.

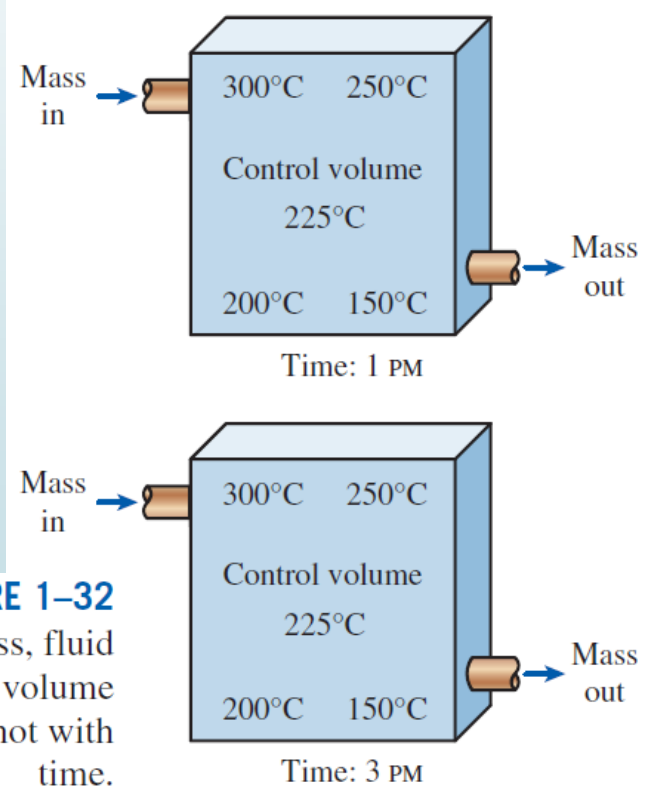


FIGURE 1–32
During a steady-flow process, fluid properties within the control volume may change with position but not with time.

The Steady Flow Process (2)

- A large number of engineering devices operate for long periods of time under the same conditions, and they are classified as *steady-flow devices*.
- Steady-flow conditions can be closely approximated by devices that are intended for continuous operation such as turbines, pumps, boilers, condensers, and heat exchangers or power plants or refrigeration systems.

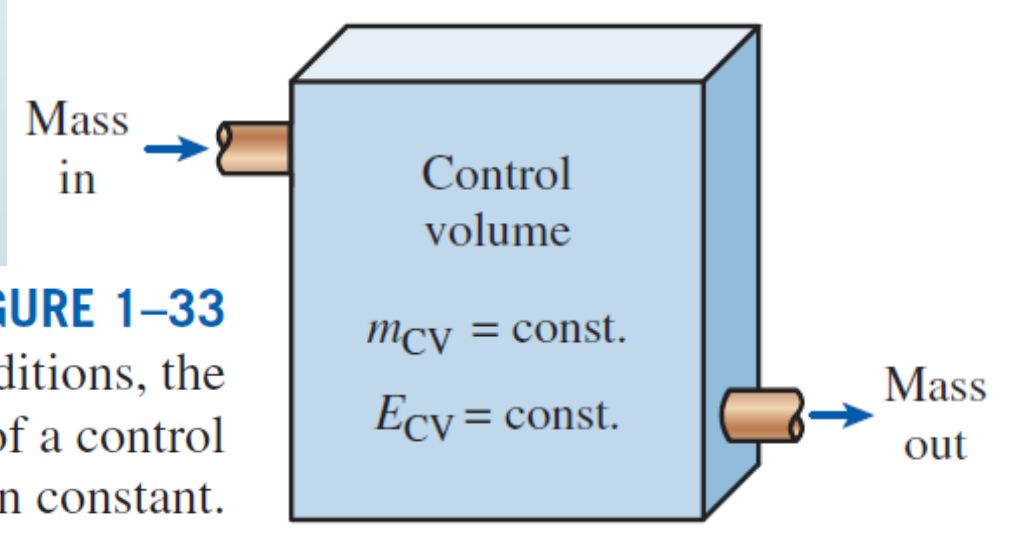


FIGURE 1–33
Under steady-flow conditions, the mass and energy contents of a control volume remain constant.

Zeroth law of thermodynamics

- **Zeroth law of thermodynamics** states that if two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other.
  - The statement seems trivial but serves as a basis for the validity of temperature measurement.
- By replacing the third body with a thermometer, the zeroth law can be restated as two bodies are in thermal equilibrium if both have the same temperature reading even if they are not in contact.

- Its value as a fundamental physical principle was recognized more than half a century after the formulation of the first and the second laws of thermodynamics.
- It was named the zeroth law since it should have preceded the first and the second laws of thermodynamics.

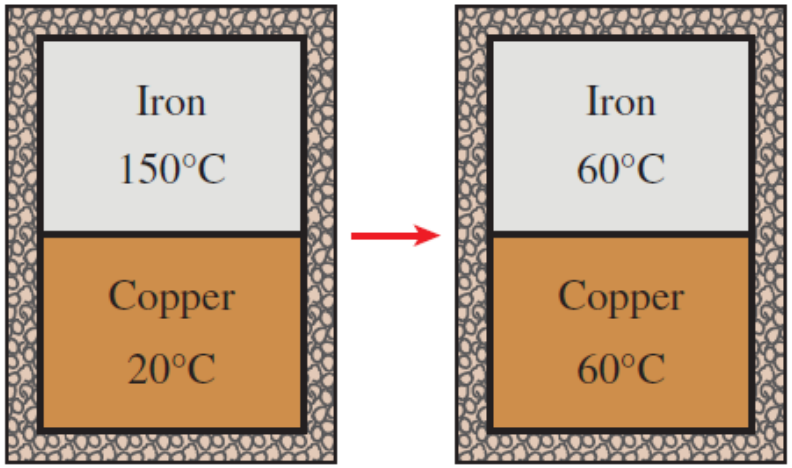


FIGURE 1–34
Two bodies reaching thermal equilibrium after being brought into contact in an isolated enclosure.

Temperature Scales

- All temperature scales are based on some easily reproducible states such as the freezing and boiling points of water: the *ice point* and the *steam point.*
- **Ice point**: A mixture of ice and water that is in equilibrium with air saturated with vapor at 1 atm pressure (0°C or 32°F).
- **Steam point**: A mixture of liquid water and water vapor (with no air) in equilibrium at 1 atm pressure (100°C or 212°F).
- **Celsius scale**: in SI unit system
- **Fahrenheit scale**: in English unit system
- **Thermodynamic temperature scale**: A temperature scale that is independent of the properties of any substance.
- **Kelvin scale** (SI) **Rankine scale** (E)
- A temperature scale nearly identical to the Kelvin scale is the **ideal-gas temperature scale**. The temperatures on this scale are measured using a constant-volume gas thermometer.

Temperature Scales (2)

- The reference temperature in the original Kelvin scale was the ***ice point***, 273.15 K, which is the temperature at which water freezes (or ice melts).
- The reference point was changed to a much more precisely reproducible point, the ***triple point*** of water (the state at which all three phases of water coexist in equilibrium), which is assigned the value 273.16 K.

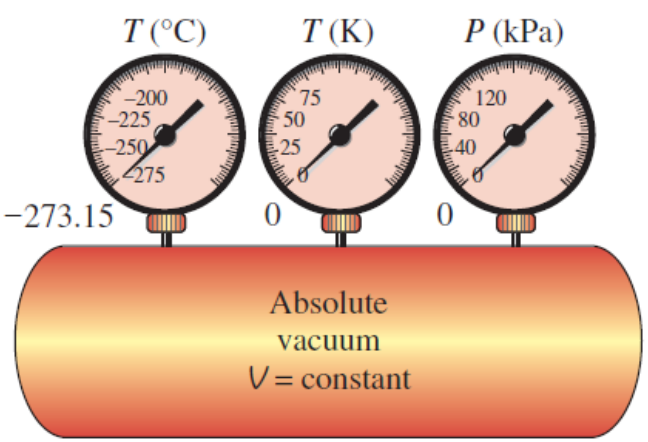


FIGURE 1–36
A constant-volume gas thermometer would read −273.15°C at absolute zero pressure.

$$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$$

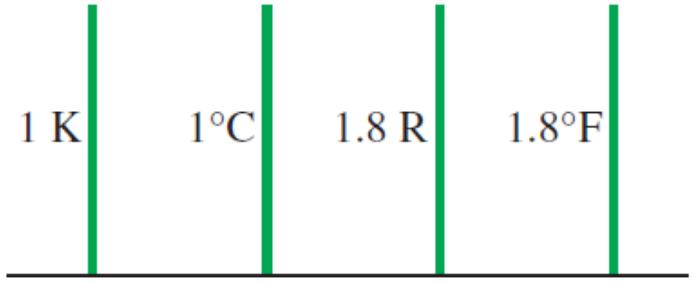
$$T(\mathrm{R}) = T(^\circ\mathrm{F}) + 459.67$$

$$T(\mathrm{R}) = 1.8T(\mathrm{K})$$

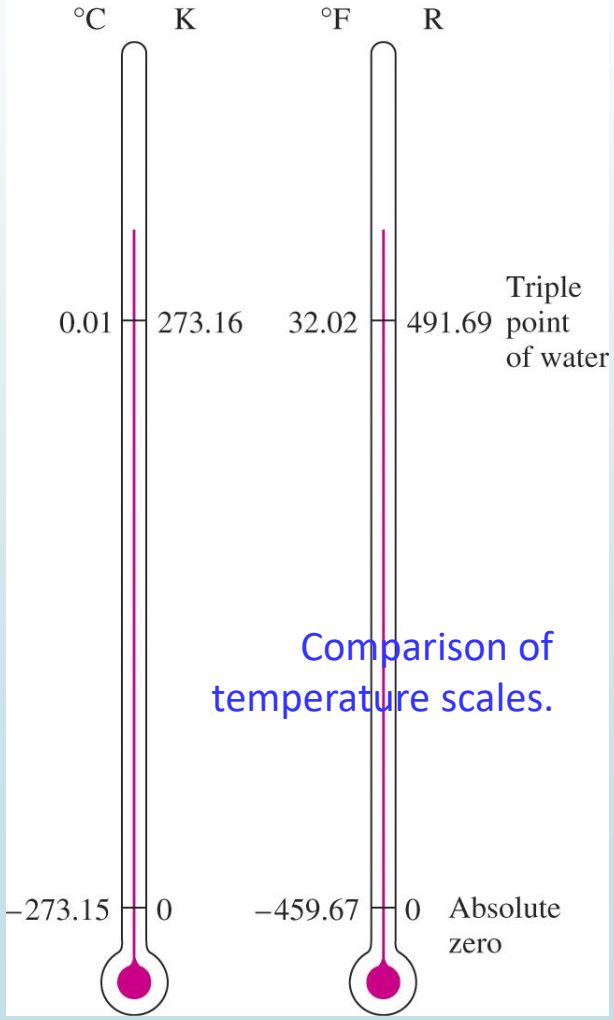
$$T(^\circ\mathrm{F}) = 1.8T(^\circ\mathrm{C}) + 32$$

$$\Delta T(\mathrm{K}) = \Delta T(^\circ\mathrm{C})$$

$$\Delta T(\mathrm{R}) = \Delta T(^\circ\mathrm{F})$$



Comparison of magnitudes of various temperature units.



Comparison of temperature scales.

1.9 Pressure

- **Pressure**: A normal force exerted by a fluid per unit area

$$1\ \text{Pa} = 1\ \text{N/m}^2$$

$$1\ \text{bar} = 10^5\ \text{Pa} = 0.1\ \text{MPa} = 100\ \text{kPa}$$

$$1\ \text{atm} = 101{,}325\ \text{Pa} = 101.325\ \text{kPa} = 1.01325\ \text{bars}$$

$$1\ \text{kgf/cm}^2 = 9.807\ \text{N/cm}^2 = 9.807 \times 10^4\ \text{N/m}^2 = 9.807 \times 10^4\ \text{Pa}$$

$$= 0.9807\ \text{bar}$$

$$= 0.9679\ \text{atm}$$

Some basic pressure gages.

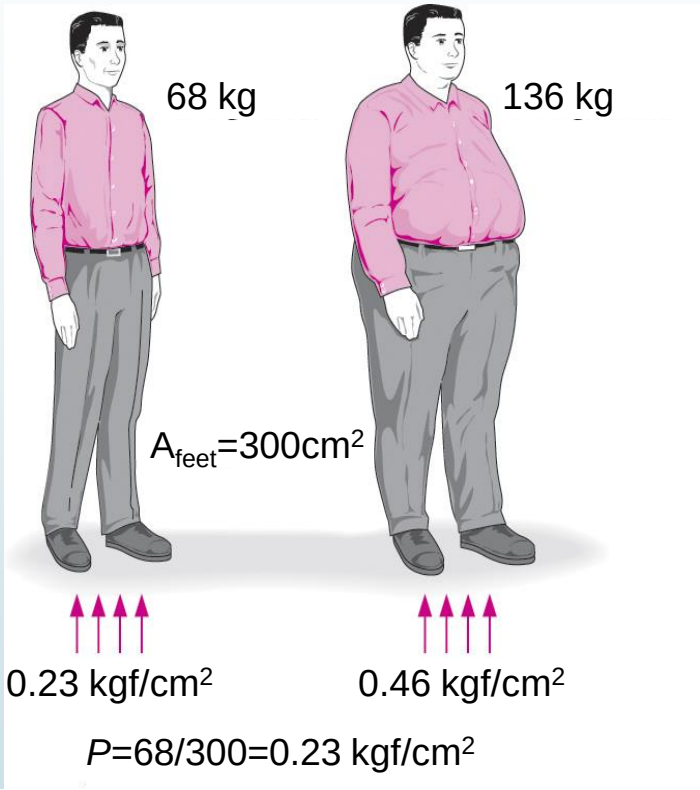


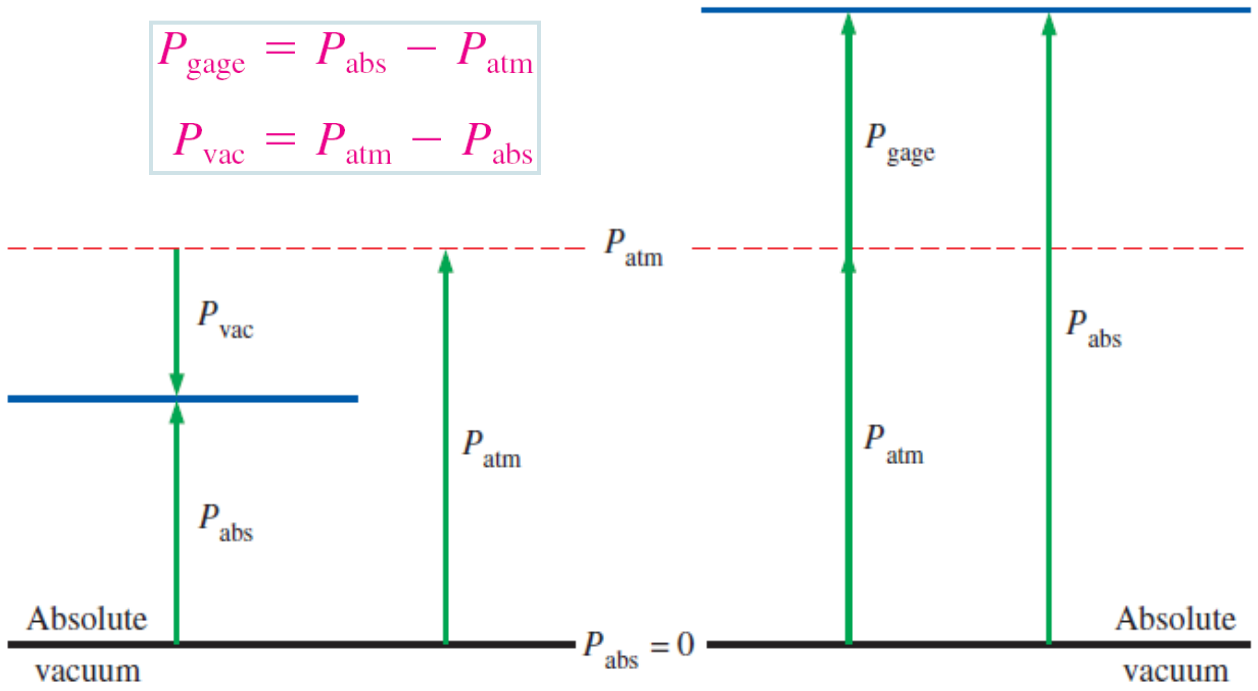
FIGURE 1–39
The normal stress (or "pressure") on the feet of a chubby person is much greater than on the feet of a slim person.

1.9 Pressure (2)

- **Absolute pressure**: The actual pressure at a given position. It is measured relative to absolute vacuum (i.e., absolute zero pressure).
- **Gage pressure**: The difference between the absolute pressure and the local atmospheric pressure. Most pressure-measuring devices are calibrated to read zero in the atmosphere, and so they indicate gage pressure.
- **Vacuum pressures**: Pressures below atmospheric pressure.

$$P_{gage} = P_{abs} - P_{atm}$$

$$P_{vac} = P_{atm} - P_{abs}$$



Throughout this text, the pressure ***P*** will denote ***absolute pressure*** unless specified otherwise.

1.9 Pressure (3)

- Variation of Pressure with Depth

$$\Delta P = P_2 - P_1 = \rho g \, \Delta z = \gamma_s \, \Delta z$$

$$P = P_{atm} + \rho g h \quad \text{or} \quad P_{gage} = \rho g h$$

- When the variation of density with elevation is known

$$\Delta P = P_2 - P_1 = -\int_1^2 \rho g \, dz$$

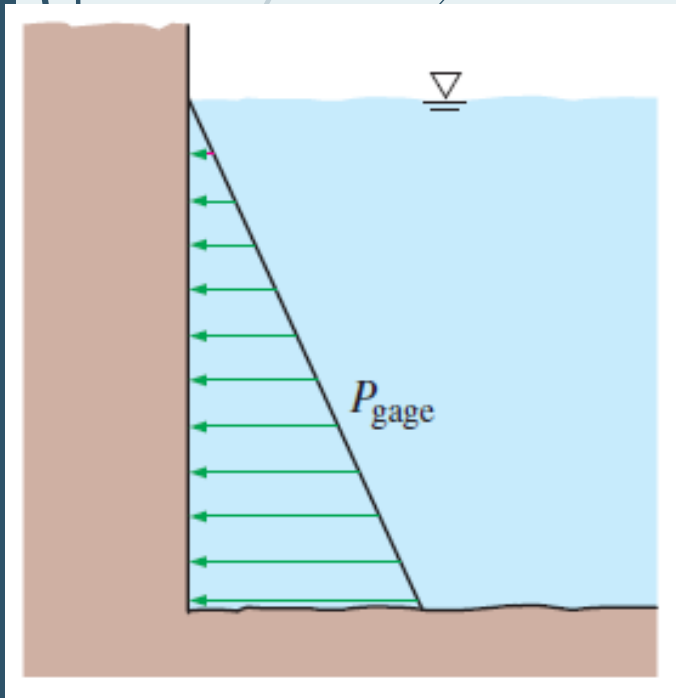


FIGURE 1–42
The pressure of a fluid at rest increases with depth (as a result of added weight).

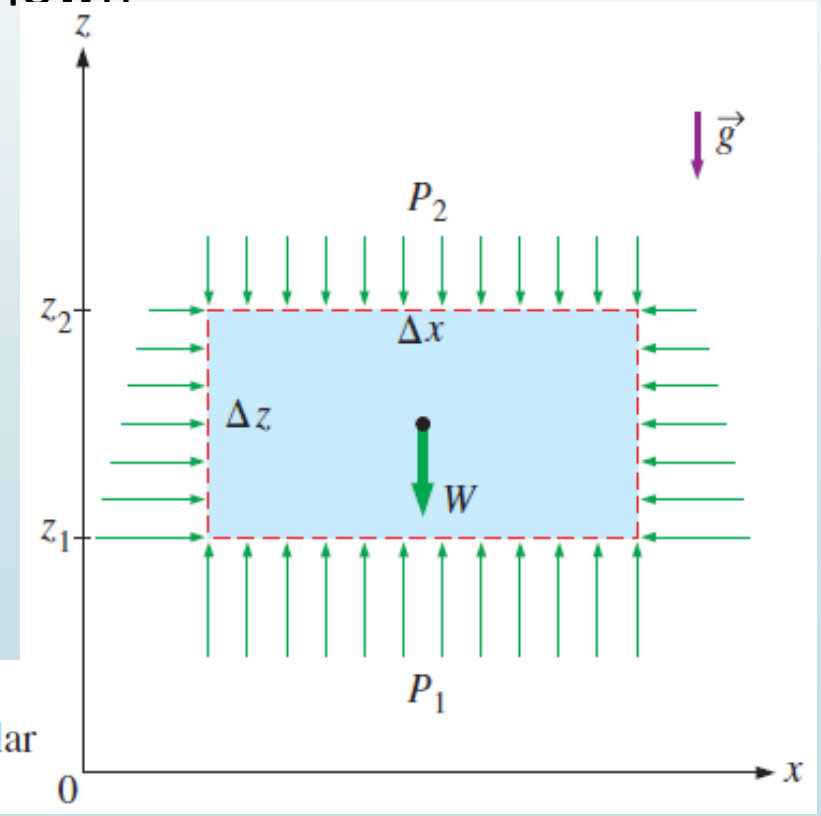
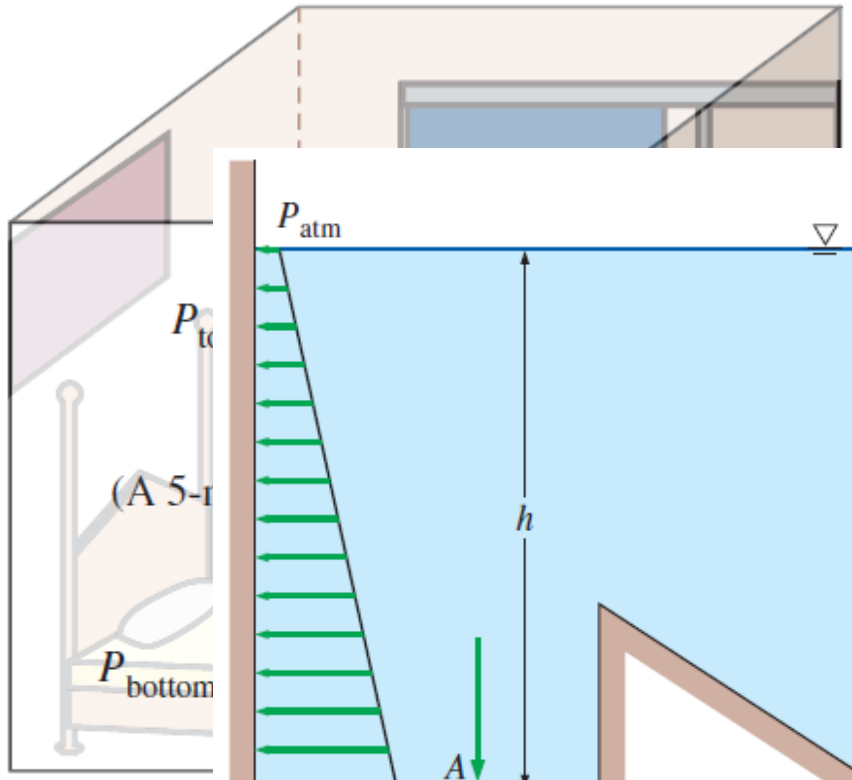


FIGURE 1–43
Free-body diagram of a rectangular fluid element in equilibrium.

1.9 Pressure (4)



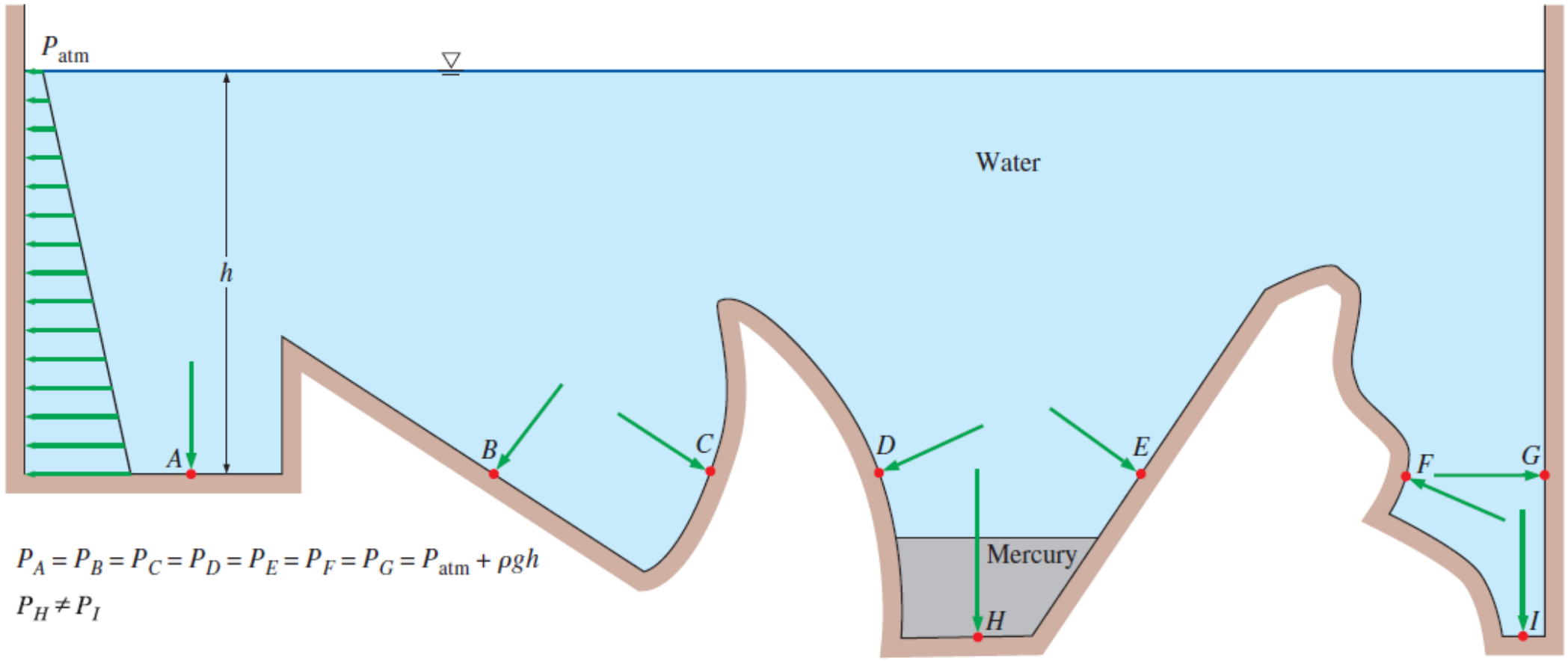


FIGURE 1–46
Under hydrostatic conditions, the pressure is the same at all points on a horizontal plane in a given fluid regardless of geometry, provided that the points are interconnected by the same fluid.

1.9 Pressure (5)

- **Pascal's law:** The pressure applied to a confined fluid increases the pressure throughout by the same amount.

$$P_1 = P_2 \quad \rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \rightarrow \quad \frac{F_2}{F_1} = \frac{A_2}{A_1}$$

- The area ratio A_2/A_1 is called the ideal mechanical advantage of the hydraulic lift.

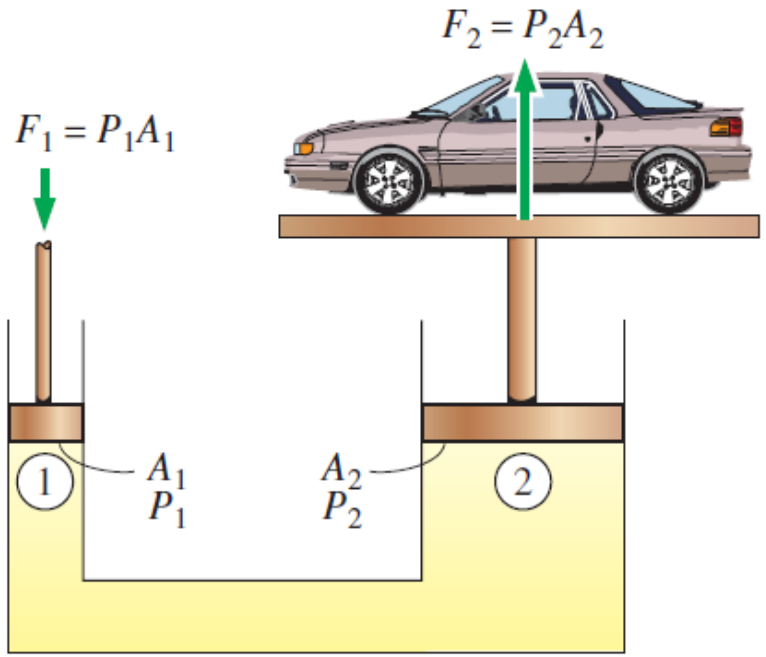


FIGURE 1–47
Lifting of a large weight by a small force by the application of Pascal's law. A common example is a hydraulic jack.

1.10 Pressure Measurement Devices

The Barometer

- Atmospheric pressure is measured by a device called a **barometer**; thus, the atmospheric pressure is often referred to as the ***barometric pressure***.

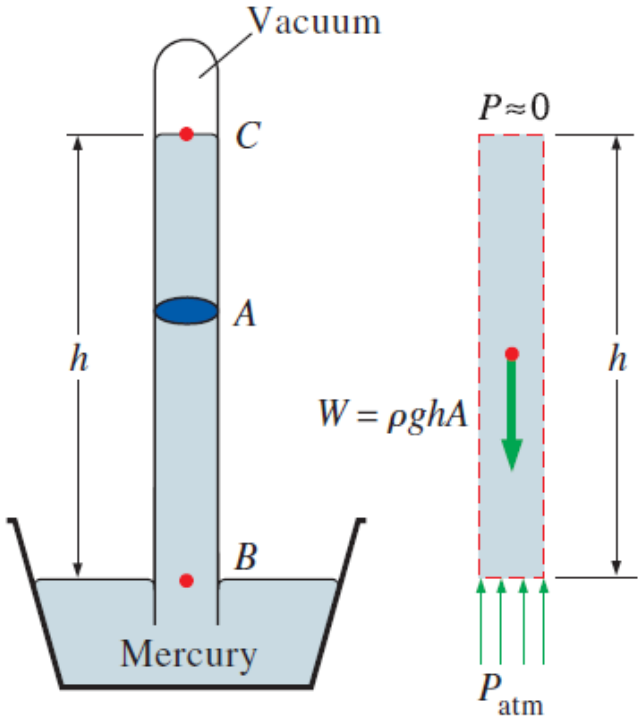


FIGURE 1–48
The basic barometer.

- A frequently used pressure unit is the *standard atmosphere*, which is defined as the pressure produced by a column of mercury 760 mm in height at 0°C (ρ_{Hg} = 13,595 kg/m³) under standard gravitational acceleration (g = 9.807 m/s²).

$$P_{atm} = \rho g h$$

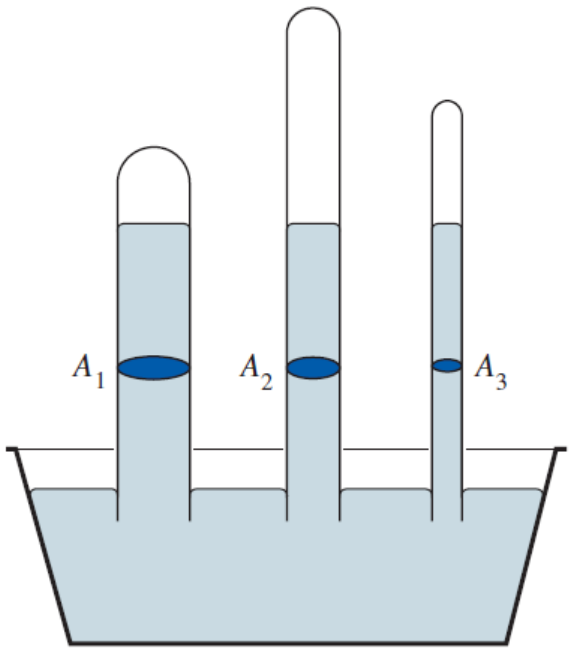
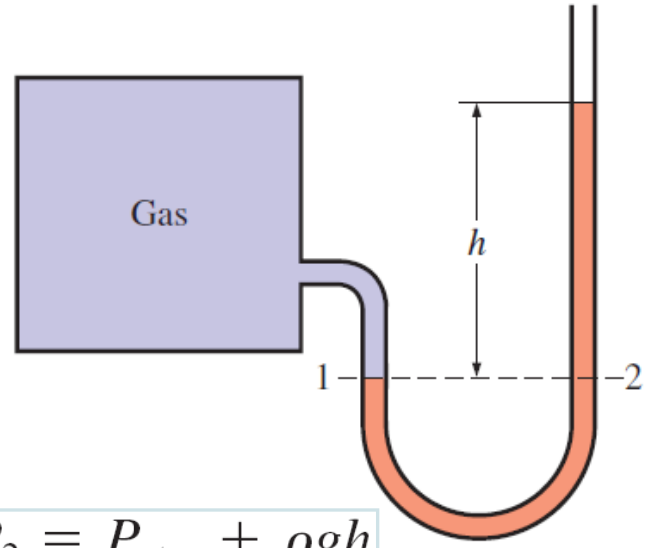


FIGURE 1–49
The length or the cross-sectional area of the tube has no effect on the height of the fluid column of a barometer, provided that the tube diameter is large enough to avoid surface tension (capillary) effects.

1.10 Pressure Measurement Devices (2)

The Manometer

- It is commonly used to measure small and moderate pressure differences. A manometer contains one or more fluids such as mercury, water, alcohol, or oil.



$$P_2 = P_{atm} + \rho g h$$

FIGURE 1–55
The basic manometer.

FIGURE 1–54
A simple U-tube manometer, with high pressure applied to the right side.

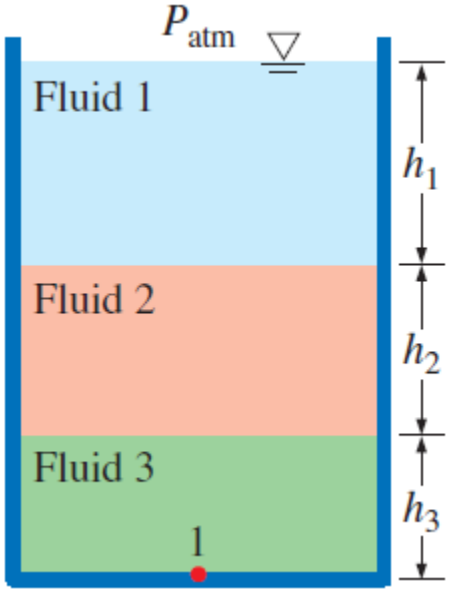


FIGURE 1–57
In stacked-up fluid layers at rest, the pressure change across each fluid layer of density ρ and height h is $\rho g h$.

$$P_{atm} + \rho_1 g h_1 + \rho_2 g h_2 + \rho_3 g h_3 = P_1$$

$$P_1 + \rho_1 g(a + h) - \rho_2 g h - \rho_1 g a = P_2$$

$$P_1 - P_2 = (\rho_2 - \rho_1) g h$$

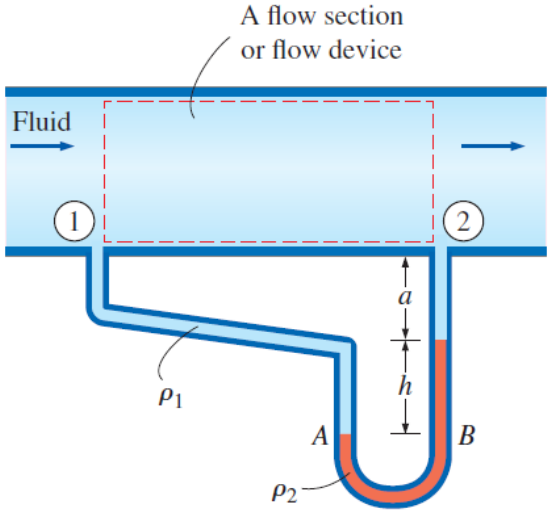


FIGURE 1–58
Measuring the pressure drop across a flow section or a flow device by a differential manometer.

1.10 Pressure Measurement Devices (3)

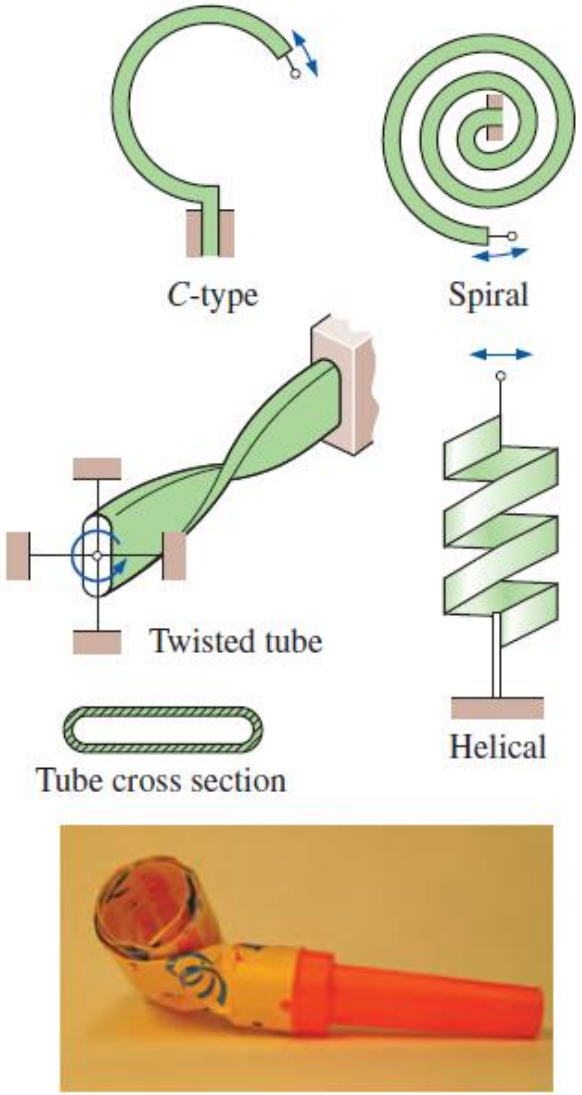


FIGURE 1–60
Various types of Bourdon tubes used to measure pressure. They work on the same principle as party noise-makers (bottom photo) due to the flat tube cross section.

Other Pressure Measurement Devices

- **Bourdon tube**: Consists of a hollow metal tube bent like a hook whose end is closed and connected to a dial indicator needle.
- **Pressure transducers**: Use various techniques to convert the pressure effect to an electrical effect such as a change in voltage, resistance, or capacitance.
  - Pressure transducers are smaller and faster, and they can be more sensitive, reliable, and precise than their mechanical counterparts.
- **Strain-gage pressure transducers**: Work by having a diaphragm deflect between two chambers open to the pressure inputs.
- **Piezoelectric transducers**: Also called solid-state pressure transducers, work on the principle that an electric potential is generated in a crystalline substance when it is subjected to mechanical pressure.

1.11 Problem-Solving Technique

By using a step-by-step approach, an engineer can reduce the solution of a complicated problem into the solution of a series of simple problems.

Step 1: Problem Statement

Briefly state the problem in your own words.

Step 2: Schematic

Draw a realistic sketch of the physical system involved, and list the relevant information on the figure.

Step 3: Assumptions and Approximations

State any appropriate assumptions and approximations made to simplify the problem to make it possible to obtain a solution.

Step 4: Physical Laws

Apply all the relevant basic physical laws and principles (such as the conservation of mass), and reduce them to their simplest form by utilizing the assumptions made.

1.11 Problem-Solving Technique (2)

Step 5: Properties

Determine the unknown properties at known states necessary to solve the problem from property relations or tables.

Step 6: Calculations

Substitute the known quantities into the simplified relations and perform the calculations to determine the unknowns.

Step 7: Reasoning, Verification, and Discussion

Check to make sure that the results obtained are reasonable and intuitive, and verify the validity of the questionable assumptions. Repeat the calculations that resulted in unreasonable values.

Thank You